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400G/lane for Linear-drive Optics A

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Abstract: We demonstrate performances of 400G/lane with a high bar lithium niobate Mach-Zehnder modulator and practical equalizations. feasibility of 400G/lane for linear-drive optics applications in NPO/CPC
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1. Introduction

The demand for high bandwidth has rapidly increased to support AI/ML applicat AI clusters. 1.6T modules utilizing 200G/lane are currently under development with This progression has driven significant research interest in 3.2T or high implementations, where novel technologies must simultaneously address per constraints, and scalability challenges [1-4]. High symbol rate modulation employ technologies, e.g., thin-film lithium niobate (TFLN) based Mach-Zehnder modulat based electro-absorption modulated lasers (EML), and heterogeneously integrate (EAM) on silicon waveguides, has been reported for 400G/lane implemen demonstrations face critical limitations for short reach connections: failure either error correction (FEC) overhead (OH), or to rely on offline digital signal proce intensive equalization, or to require of high FEC BER thresholds (e.g. $2.4\text{E-}2$ wi for 200G/lane systems adopt KP4 FEC and a concatenated KP4 and Hammin connections for power efficiency and low latency [4,5], scalable to 400G/lane wi and $4.85\text{E-}3$ (KP4 + Hamming), corresponding to gross bitrates of 425Gb/s and 4

Linear-drive optics (LDO) eliminates digital signal processing (DSP) in requirements of AI/ML applications such as low power, low latency and low co form factors: one is Linear-drive Pluggable Optics (LPO), and the other is linear Package Optics (NPO). LPO modules show promise in adapting to links with exist however, achieving 400G/lane in LPO is impossible in conventional models due switch [4]. To support 400G/lane, new high-density RF connectors are proposed that 3dB bandwidth (BW) can be over 120GHz [6], to enable 400G/lane with the

In this paper, we demonstrate 400G/lane solutions, compliant with KP4 FE specifications, utilizing a high bandwidth efficient TFLN MZM with practical equa applications. We experimentally demonstrated performances of 160Gbd PAM4 with KP4 + Hamming FEC using 15 feed-forward equalizer (FFE) taps. The s